

A MATHEMATICIAN'S APOLOGY
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1 Doers versus Talkers

The function of a mathematician is to do something, to prove new theorems, to add to mathematics, and not to talk about what he or other mathematicians have done. Exposition, criticism, appreciation, is work for second-rate minds

Would the life of the best of critics really have seemed to be comparable with that of a scholar and a poet?

I write about mathematics because, like any other mathematician who has passed sixty, I have no longer the freshness of mind, the energy, or the patience to carry on effectively with my proper job.

2 Pure versus Applied

It is probable, since the sensational triumphs of Einstein, that stellar astronomy and atomic physics are the only sciences which stand higher in popular estimation (than mathematics).

The mass of mathematical truth is obvious and imposing; its practical applications, the bridges and steam-engines and dynamos, obtrude themselves on the dullest imagination. The public does not need to be convinced that there is something in mathematics.

Any genuine mathematician must feel that it is not on these crude achievements that the real case for mathematics rests, that the popular reputation of mathematics is based largely on ignorance and confusion, and that there is room for a more rational defense

Why is it really worth while to make a serious study of mathematics?
What is the proper justification of a mathematician's life?

3 The standard apology

I do what I do because it is the one and only thing that I can do well. I agree that it might be better to be a poet or a mathematician, but unfortunately I have no talent for such pursuits

Most people can do nothing at all well. Perhaps 5 or 10 percent of men can do something rather well. It is a tiny minority who can do anything *really* well, and the number of men who can do two things well is negligible.

We have to take account of the differences in value between different activities. Yet it is seldom that such differences of value will turn the scale in a man's choice of a career, which will almost always be dictated by the limitations of his natural abilities.

If a man is in any sense a real mathematician, then it is a hundred to one that his mathematics will be far better than anything else he can do, and that he would be silly if he surrendered any decent opportunity of exercising his one talent in order to do undistinguished work in other fields.

4 Young versus Old

No mathematician should ever allow himself to forget that mathematics, more than any other art or science, is a young man's game.

Newton gave up mathematics at fifty, and lost his enthusiasm long before. Galois died at twenty-one, Abel at twenty-seven, Ramanujan at thirty-three, Riemann at forty. I do not know of a major mathematical advance initiated by a man past fifty.

It is very hard to find an instance of a first-rate mathematician who has abandoned mathematics and has attained first-rate distinction in any other field.

Every young mathematician of real talent whom I have known has been faithful to mathematics, and not from lack of ambition but from an abundance of it; they have all recognized that there, if anywhere, lay the road to a life of any distinction

5 The Humbler Apology

There is *nothing* that I can do particularly well. I do what I do because it came my way. I never really had a chance of doing anything else.

It is quite true that most people can do nothing well. If so, it matters very little what career they choose, and there is really nothing more to say about it

6 Rationalization

Is mathematics, what I and other mathematicians mean by mathematics, worth doing; and if so, why? The study of mathematics is, if unprofitable, a perfectly harmless and innocent occupation

Questions temporarily left open: Is mathematics “unprofitable?” Is mathematics ‘useful?’ Is mathematics ‘harmless?’

The scale of the universe is large and, if we are wasting our time, the waste of the lives of a few university dons is no such overwhelming catastrophe. I am assuming that we dons really had our little talents, and that we could hardly be wrong if we did our best to cultivate them fully

What we do may be small, but it has a certain character of permanence; and to have produced anything of the slightest permanent interest, whether it be a copy of verse or a geometrical theorem, is to have done something utterly beyond the powers of the vast majority of men.

7 Identical Motives

A man's first duty, a young man's at any rate, is to be ambitious. The noblest ambition is that of leaving behind one something of permanent value

Ambition has been the driving force behind nearly all the best work of the world. Practically all substantial contributions to human happiness have been made by ambitious men

We must guard against a common fallacy of supposing that the men whose work most benefits humanity are thinking much of that while they do it. A physiologist may indeed be glad to remember that his work will benefit mankind, but the motives which provide the force and the inspiration for it are indistinguishable from those of a classical scholar or a mathematician.

The three most important motives which lead men to prosecute research are **intellectual curiosity**, **professional pride** and **ambition**, and no decent man need be ashamed of this.